

接下来计算 $\hat{U}_0^\dagger \hat{H} \hat{U}_0$ 。利用

$$\hat{U}_0^\dagger \hat{S}_z \hat{U}_0 = \hat{S}_z, \quad (18)$$

$$\hat{U}_0^\dagger \hat{S}_x \hat{U}_0 = \hat{S}_x \cos(\omega t) - \hat{S}_y \sin(\omega t), \quad (19)$$

$$\hat{U}_0^\dagger \hat{S}_y \hat{U}_0 = \hat{S}_x \sin(\omega t) + \hat{S}_y \cos(\omega t). \quad (20)$$

将实验室哈密顿量 (6) 变换：

$$\begin{aligned} \hat{U}_0^\dagger \hat{H} \hat{U}_0 &= -\gamma B_0 \hat{S}_z - \gamma B_1 \left(\hat{U}_0^\dagger \hat{S}_x \hat{U}_0 \cos(\omega t) - \hat{U}_0^\dagger \hat{S}_y \hat{U}_0 \sin(\omega t) \right) \\ &= -\gamma B_0 \hat{S}_z - \gamma B_1 \left[\left(\hat{S}_x \cos(\omega t) - \hat{S}_y \sin(\omega t) \right) \cos(\omega t) \right. \\ &\quad \left. - \left(\hat{S}_x \sin(\omega t) + \hat{S}_y \cos(\omega t) \right) \sin(\omega t) \right] \\ &= -\gamma B_0 \hat{S}_z - \gamma B_1 \left[\hat{S}_x (\cos^2 \omega t + \sin^2 \omega t) - \hat{S}_y (\sin \omega t \cos \omega t - \cos \omega t \sin \omega t) \right] \\ &= -\gamma B_0 \hat{S}_z - \gamma B_1 \hat{S}_x. \end{aligned} \quad (21)$$

因此旋转坐标系中的有效哈密顿量为

$$\hat{H}_{\text{rot}} = \left(-\gamma B_0 \hat{S}_z - \gamma B_1 \hat{S}_x \right) - \omega \hat{S}_z = -(\gamma B_0 + \omega) \hat{S}_z - \gamma B_1 \hat{S}_x. \quad (22)$$

在共振条件 $\omega = -\gamma B_0 \equiv \omega_0$ 下，第一项消失：

$$\boxed{\hat{H}_{\text{rot}} = -\gamma B_1 \hat{S}_x = -\frac{\gamma \hbar B_1}{2} \sigma_x}. \quad (23)$$

这是一个与时间无关的哈密顿量。

1.3 时间演化算符与翻转角

旋转坐标系中的时间演化算符为

$$\hat{U}_{\text{rot}}(t) = \exp\left(-\frac{i}{\hbar} \hat{H}_{\text{rot}} t\right) = \exp\left(i \frac{\gamma B_1 t}{2} \sigma_x\right). \quad (24)$$

定义翻转角

$$\theta = \gamma B_1 t, \quad \text{则} \quad \hat{U}_{\text{rot}}(t) = \exp\left(i \frac{\theta}{2} \sigma_x\right) = \cos \frac{\theta}{2} I + i \sin \frac{\theta}{2} \sigma_x. \quad (25)$$

1.4 对单个自旋态的作用

初始时刻（热平衡）自旋处于 $+z$ 方向的本征态 $|\uparrow_z\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ ，则

$$\begin{aligned} |\psi_{\text{rot}}(t)\rangle &= \hat{U}_{\text{rot}}(t) |\uparrow_z\rangle \\ &= \left(\cos \frac{\theta}{2} I + i \sin \frac{\theta}{2} \sigma_x \right) |\uparrow_z\rangle \\ &= \cos \frac{\theta}{2} |\uparrow_z\rangle + i \sin \frac{\theta}{2} |\downarrow_z\rangle. \end{aligned} \quad (26)$$